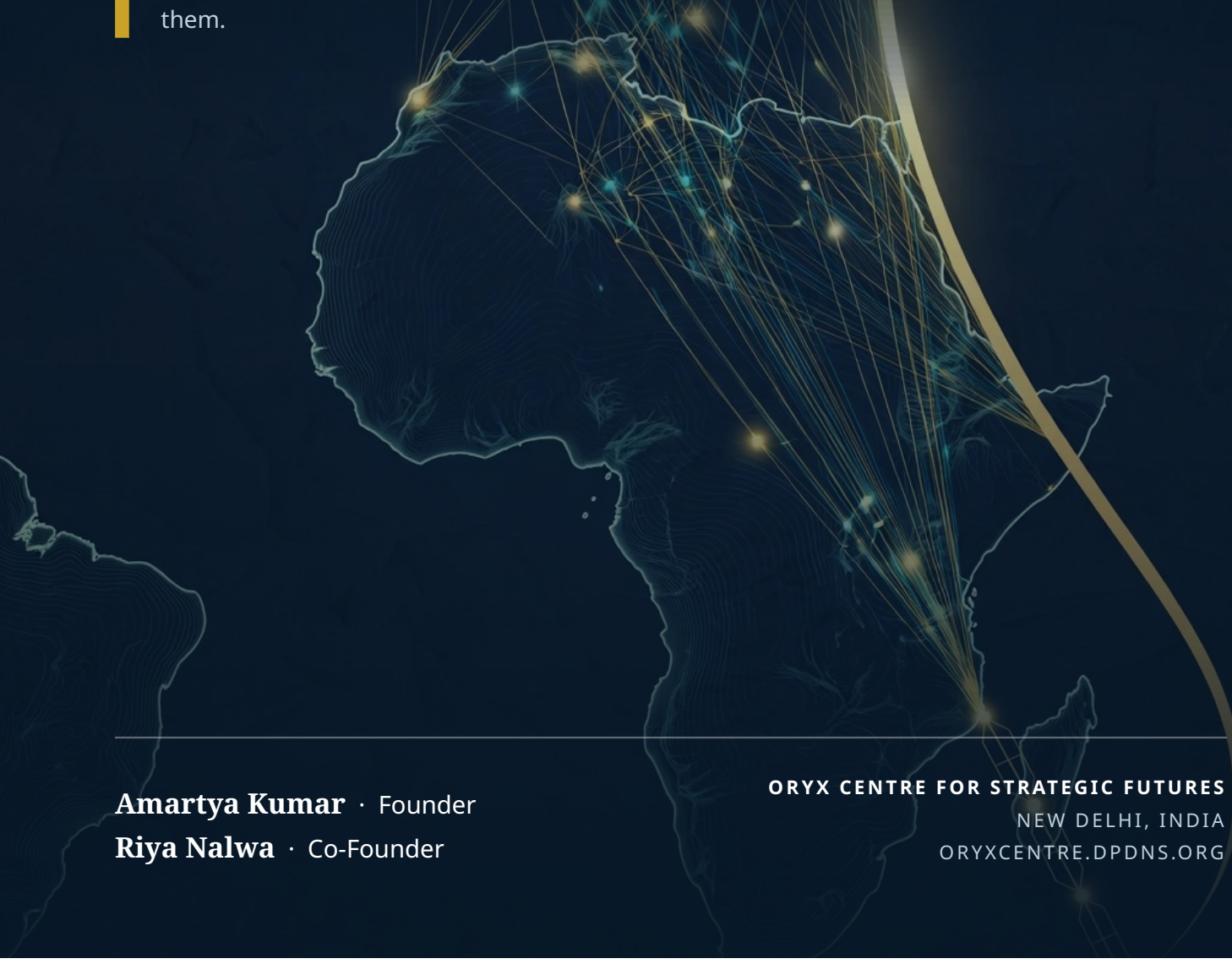
ORYX
CENTRE FOR STRATEGIC FUTURES

WORKING PAPER · OCTOBER 2026

From Answers to Auditable Intelligence

A Governance-First Framework for Agentic AI in Strategic Research and Public Policy

Agentic AI can draft a persuasive answer in seconds. Strategic research, however, is not short of answers — it is short of answers whose evidence can be inspected, whose uncertainty is visible, and whose conclusions a named human being is willing to own. This paper designs the workflow that produces them.



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About this paper

A working paper proposing a six-layer governance framework and an evaluation protocol for agentic AI used in strategic research — written to be tested, not merely cited, and updated in October 2026 with a first published measurement of the problem it addresses and a first implementation verification run.

RESEARCH STATUS — READ FIRST. This is a working paper, not a peer-reviewed publication. The framework it proposes has **not been validated** by a blinded pilot or independent review, and **no public product** has been deployed. The Oryx Centre for Strategic Futures is an early-stage, non-profit research and policy initiative based in New Delhi, India, and is not yet formally registered in any jurisdiction. Two things have changed since v2.0, and both are labelled where they appear: **(1)** a companion empirical study by the authors has been published, measuring citation failure in AI-generated strategic research briefs (Kumar & Nalwa, 2026, cited in §2); **(2)** the framework has been implemented as an internal minimum viable product with a first end-to-end verification run on a single test question (§8.4) — an engineering check, not the blinded partner pilot of §7–§8, which still awaits a partner. Every other number in this paper remains a protocol target, a design threshold, or a clearly labelled illustrative example. No institution named here is a partner, funder, or endorser of Oryx unless a written agreement says so.

Field	Detail
Title	From Answers to Auditable Intelligence: A Governance-First Framework for Agentic AI in Strategic Research and Public Policy
Authors	Amartya Kumar (Founder) and Riya Nalwa (Co-Founder), Oryx Centre for Strategic Futures
Publisher	Oryx Centre for Strategic Futures, New Delhi, India — oryxcentre.dpdns.org
Date / version	October 2026 · Version 3.0 (updated edition: adds a published companion measurement and a first implementation verification run; the framework and the evaluation protocol are unchanged from v2.0)
Contact	amartya@oryxcentre.dpdns.org
Keywords	Agentic AI; strategic foresight; research automation; public policy; AI governance; evidence traceability; uncertainty calibration; Global South
Intellectual basis	A focused narrative synthesis of published literature and public guidance (Wang et al. 2024; Guo et al. 2024; NIST AI RMF 1.0, 2023; OECD AI Principles, 2024; UNESCO guidance, Miao & Holmes 2023). Full references on the final pages.

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EXECUTIVE SUMMARY

The argument in brief

Judge agentic AI for research not by how fluently it answers, but by whether every material claim it produces can be traced, challenged, and owned.

The problem. Agentic AI systems can plan multi-step tasks, use tools, retrieve information, and coordinate intermediate outputs across specialised components. For research and policy teams facing more information than they can assess, that capability is genuinely valuable. It is also dangerous in a specific, under-appreciated way: the failure mode of a modern research agent is rarely an obvious error. It is a *plausible* output — fluent, well-structured, confidently cited — whose evidence does not in fact support its claims. A citation may not support the sentence attached to it. A source may be outdated, unrepresentative, or circular. Several agents may repeat the same initial mistake in different words, manufacturing the appearance of corroboration. A smooth synthesis may quietly erase genuine disagreement among experts. In low-stakes content production these are quality issues. In research that informs public policy, investment, technology governance, or geopolitical judgement, they are decision risks.

The proposal. This paper proposes a governance-first framework that treats an agentic research system as an *auditable process* rather than an answer generator. It has six layers: **(1) question framing**, in which a human research lead fixes the decision context, scope, and evidentiary bar before any model runs; **(2) evidence acquisition**, in which sources are gathered and qualified with full provenance; **(3) claim-level traceability**, in which draft output is decomposed into material claims, each linked to the passages that support, contextualise, or contradict it; **(4) adversarial review**, a separate pass whose job is to attack the draft — hunting counter-evidence, missing perspectives, and overconfident language; **(5) human decision gates**, at which named, accountable reviewers approve the question, inspect material claims, and authorise release, with rigour proportionate to consequence; and **(6) monitoring and correction**, which versions outputs, records failure modes, and converts each error into an organisational lesson. Human accountability is not one layer among six: it runs across all of them.

The test. A framework that cannot fail a test is a slogan. The paper therefore specifies an evaluation protocol: a bounded pilot comparing the framework-guided workflow against a conventional researcher-led baseline on the same question, under the same source-access conditions, with outputs scored by reviewers who do not know which workflow produced them. Performance is measured on sourced-claim rate, citation precision and recall, evidence coverage, fabrication rate, uncertainty calibration, reproducibility, reviewer effort, and human-override outcomes — with thresholds fixed *before* the pilot begins. A workflow that is faster but harder to verify, thinner in coverage, or more likely to be waved through uncritically has negative net value, and the protocol is designed to say so.

The context. The framework is designed from, and for, contexts where the stakes are highest and governance capacity thinnest — including research institutions across the Global South, where evidence is scattered across languages and unevenly indexed sources, and where a tool that cannot be audited may cost more than it saves. Section 9 develops this dimension; Section 11 sets out an 18-month roadmap to a first independently reviewed pilot; Section 12 states plainly what this paper does not know.

> *"In high-consequence research, the value of agentic AI should be judged not by fluency or speed alone, but by whether it improves the production of auditable, contestable, and decision-useful analysis under human accountability."*

HOW TO READ THE NUMBERS IN THIS PAPER

Four kinds of numbers appear, and they are always labelled. **TARGET** a threshold the evaluation protocol proposes to require — a design commitment, not a result. **ILLUSTRATIVE** an invented example used to show how a ledger, score, or workflow would look in practice. **MEASURED** a result from an Oryx study or verification run, with its sample and date stated where it appears. Anything else is a verifiable fact about the cited literature or public guidance. No number in this paper reports a measurement of the

framework's performance against its §7 targets, because the blinded pilot that would produce such a measurement has not yet run.

SECTION 1

1 Introduction: research has an auditability gap, not an answer gap

The scarce resource in strategic research is no longer text. It is verified, contestable, decision-ready judgement.

Research and policy organisations face a structural mismatch between the volume of available information and the time available to assess it. Strategic questions — how an emerging technology may affect national capabilities, markets, institutions, or international relations — require researchers to assemble fragmented evidence, compare competing interpretations, identify uncertainty, and communicate conclusions to decision-makers who will act on them. The evidence is heterogeneous (official documents, academic research, company statements, datasets, journalism, expert commentary, local knowledge), the conclusions are often inferential, and the cost of error is asymmetric: a mistaken date in a background note is a nuisance; an unsupported claim about a government's policy, a security risk, or a company's financial position can move real decisions.

Large language models already assist with summarisation, classification, drafting, and question answering. Agentic systems extend this by organising multi-step work: planning, calling tools, maintaining intermediate state, and delegating subtasks to specialised components. Surveys of LLM-based autonomous agents and multi-agent systems document rapid architectural progress — and, consistently, open challenges around reliability, evaluation, and control (Wang et al., 2024; Guo et al., 2024). The capability curve is steep. The auditability curve is not.

Consider what a reviewer actually receives from a typical agentic research tool: a polished narrative, with citations. To trust it, the reviewer must, for every material claim, find the source, locate the passage, and judge whether it supports the claim as written — then repeat for the claims the narrative *omitted*. Verification effort scales with output length, so the time saved in drafting is quietly repaid in checking — or, worse, not repaid at all, because fluency discourages checking. The bottleneck has not been removed; it has been moved downstream and made less visible. This is the auditability gap: production of research-shaped text has become cheap, while production of *verifiable* research has not.

1.1 The proposition

This paper advances a simple proposition: **agentic AI should be designed as an auditable research process, not merely as an answer generator.** Concretely, that means the workflow — not just the final text — is the product. A governance-first workflow keeps the provenance of evidence visible at every step, decomposes output into claims whose support can be individually inspected, makes uncertainty explicit rather than smoothing it away, separates factual claims from interpretation and recommendation, institutionalises an adversarial challenge pass, and reserves consequential judgements for accountable human reviewers whose approvals are recorded. It also treats evaluation as a design component, not an afterthought: the framework ships with the protocol by which it can be shown to fail.

1.2 What this paper is, and is not

This is a conceptual working paper from an early-stage research initiative. It synthesises published literature and public governance guidance into an operational design, expands that design into implementation detail, and specifies how it should be tested. It is not an empirical study, a product announcement, or a claim of measured performance. Its intended readers are research directors, policy-analysis teams, funders of research infrastructure, and the governance community — anyone deciding whether, and under what controls, agentic AI belongs inside a research process whose outputs carry consequences.

SECTION 2

2 The problem landscape: five ways plausible output fails

Each failure mode below is quiet, compounding, and specifically dangerous when the reader is a decision-maker.

Plausibility is the core hazard. A hallucinated fact in a chatbot conversation is an annoyance; the same fact, laundered through a three-agent research pipeline into a ministerial brief, is a liability. Five failure modes recur across the literature and across practitioner experience with LLM-supported research. They are worth naming precisely, because the framework in Section 5 exists to counter each of them — and because any evaluation that does not probe for them is measuring the wrong thing.

2.1 Fabrication and false attribution

The best-known failure: the system asserts something no source says, or attaches a real citation to a claim the source does not make. Citation presence creates an unwarranted sense of verification — readers check *that* a source exists far more readily than *whether it supports the sentence*. In multi-step agentic workflows the risk compounds: an early unsupported claim becomes an input to later steps, acquiring downstream elaboration that makes it look increasingly established.

2.2 Coverage illusion

Retrieval systems surface what is indexed, recent, English-language, and institutionally visible. A synthesis built on that surface can look comprehensive while missing the local-language regulation, the dissenting technical assessment, or the dataset that reframes the question. The gap is invisible precisely because nothing in the output marks where the evidence ran out.

2.3 Manufactured consensus

Multi-agent designs are often described as self-checking: one agent researches, another verifies. But agents sharing a model family, training data, and the same retrieved sources do not provide independent corroboration; they provide *correlated* agreement. Likewise, a synthesis tuned for coherence will tend to average away genuine expert disagreement, presenting a contested forecast as settled. Agreement among agents is evidence of shared inputs, not of truth.

2.4 Confidence laundering

Uncertainty present in sources — hedged findings, wide intervals, explicit caveats — is progressively stripped as material moves from source, to notes, to synthesis, to headline. Each paraphrase firms the claim up slightly. The final reader inherits a certainty the evidence never had, with no marker showing where the hedges were lost. Calibration failures of this kind are especially dangerous in foresight work, where the honest output is often a range of scenarios, not a point prediction.

2.5 Accountability evaporation

When a brief is wrong, who erred — the researcher who commissioned it, the model that drafted it, the tool that retrieved for it? Diffused responsibility produces two symmetric pathologies: automation bias, in which reviewers over-trust fluent output and under-check it; and responsibility dumping, in which the human nominal author disowns the machine's contribution. Both erode the norm on which policy research depends: that a named person can be asked *why do you believe this?* and can answer.

Table 1 — Failure-mode atlas: how plausible-but-unverifiable output fails in strategic research, and where this framework counters it.

Failure mode	How it presents	Decision risk	Countered at
Fabrication & false attribution	Fluent claim; citation exists but does not support it; error propagates through pipeline steps	Policy built on a fact that was never established	Layers 2, 3 · §7 metrics

Coverage illusion	Comprehensive-looking synthesis over a narrow, indexed, English-first evidence base	Blind spots mistaken for settled ground	Layers 1, 2 · coverage metric
Manufactured consensus	Correlated agents agree; contested questions read as settled	False confidence in forecasts and assessments	Layer 4 · calibration metric
Confidence laundering	Hedges and caveats stripped at each paraphrase; certainty inflates downstream	Decision-makers act on firmer claims than evidence warrants	Layers 3, 4 · calibration metric
Accountability evaporation	No named owner; reviewers assume someone else checked	Errors uncorrected; trust in the institution damaged	Layer 5 · override & review log

MEASURED EVIDENCE FOR THE FAILURE MODES — COMPANION STUDY, OCTOBER 2026

In a companion pilot study published after v2.0 (Kumar & Nalwa, 2026), the authors had an AI system write 10 strategic research briefs and checked every citation by hand. MEASURED Of 66 citations, 31 (47%) were broken (links that did not resolve), 17 (26%) verified, 7 (11%) mismatched the claim they were attached to, and 11 (17%) could not be verified. Only 5 citations (8%) both resolved and fully supported the claim they were cited for. Across the briefs, 49% of material claims (93 of 190) carried any citation at all. These are measurements of *unaided* AI-generated briefs — the problem this framework addresses — not measurements of the framework itself, which remains untested against §7. The study is published on Zenodo (DOI 10.5281/zenodo.23630684); a larger replication is in progress and its interim results are deliberately not cited here.

A NOTE ON ADVERSARIAL INPUTS

A sixth hazard sits beneath the five above: retrieved material is untrusted. Web pages and documents can contain instructions crafted to redirect an agent (prompt injection), and tools can be misused when permissions are broad. A research workflow that treats everything it reads as data — never as instructions — and that gates consequential actions behind human approval is not being cautious; it is being minimally competent. Section 10 returns to this.

SECTION 3

3 Scope, method and evidence base

A synthesis and a design proposal — built from published guidance, written to be falsified by a pilot.

Scope. The framework addresses agentic AI used for *strategic research*: open-ended analytical work whose outputs inform decisions in public policy, geopolitics, technology governance, investment, and institutional strategy. It does not address fully automated decision-making (which it argues against for consequential questions), creative content generation, or low-stakes summarisation, where lighter controls are proportionate. Controls in this paper are deliberately risk-tiered: the rigour of review scales with the consequence of error.

Method. The paper uses a focused narrative synthesis of public guidance and research literature on LLM-based agents, trustworthy-AI risk management, and human-centred use of generative AI in research: surveys of LLM-based autonomous agents (Wang et al., 2024) and multi-agent systems (Guo et al., 2024); the US National Institute of Standards and Technology's AI Risk Management Framework (NIST, 2023); the OECD AI Principles as updated in 2024 (OECD, 2024); and UNESCO's guidance on generative AI in education and research (Miao & Holmes, 2023). The framework translates these sources' principles into workflow controls and an evaluation design. Section 12 records the limits of this method honestly: narrative synthesis reflects the authors' selection, and no element of the framework has been validated through field deployment, controlled experiment, or independent peer review.

Design stance. Three commitments run through everything that follows. First, *provenance before prose*: evidence is registered and qualified before synthesis begins, and synthesis may not outrun the register. Second, *claims, not documents*: the unit of quality control is the individual material claim and its support relationship, not the report as a whole. Third, *humans own conclusions*: the system's role is to make human judgement faster to exercise and easier to defend — never to simulate its presence.

SECTION 4

4 Design principles — and how this differs from current practice

Governance principles only matter when they become workflow controls. This section makes the translation explicit, then positions the framework against the ways agentic research is usually deployed today.

NIST's AI Risk Management Framework treats trustworthiness as a composite — validity and reliability, safety, security, accountability and transparency, explainability, privacy, and fairness — rather than a single score (NIST, 2023). The OECD AI Principles emphasise human rights and democratic values, transparency, robustness, safety, and accountability (OECD, 2024). UNESCO's guidance stresses human agency, inclusion, privacy, and the obligation to *validate* generative AI in research rather than assume capability implies legitimacy (Miao & Holmes, 2023). Sound principles — but a principle pinned to a wall does not check a citation. The framework's first move is to convert each into a control that blocks, flags, or records something in the workflow:

Table 2 — From principle to control: how established governance guidance is operationalised in this framework.

Principle (source)	What it demands	Workflow control in this framework
Accountability & transparency (NIST; OECD)	A named human answers for each output; the basis of conclusions is inspectable	Signed review log at human gates; every material claim carries a provenance chain (Layers 3, 5)
Validity & reliability (NIST)	Outputs should be correct, and remain correct under scrutiny and repetition	Claim-level support checks; adversarial pass; reproducibility scoring (Layers 3, 4; §7)
Explainability (NIST; OECD)	Users can understand why an output says what it says	Evidence maps precede prose; claim types (fact / inference / forecast / recommendation) are labelled, not blended (Layers 2, 3)
Human agency (UNESCO; OECD)	AI supports human judgement; it does not quietly replace it	Question framing and release are human acts; system must request clarification rather than invent scope (Layers 1, 5)
Inclusion (UNESCO; OECD)	Whose evidence and perspectives are missing is a first-order question	Source-category quotas in the charter; language and geography coverage logged (Layers 1, 2; §9)
Safety & security (NIST; OECD)	Systems resist manipulation and do not leak what they hold	Retrieved content treated as untrusted data; tool permissions minimised; sensitive inputs authorised-only (Layers 2, 6; §10)

4.1 Positioning against current practice

Four patterns dominate how agentic AI is currently used for research. Each optimises something real; each leaves the auditability gap open.

Table 3 — Positioning: common deployment patterns versus this framework (typical practice, not named products).

Pattern	What it optimises	Where auditability fails	What this framework changes
Chat-with-citations	Speed of a first answer		

		Citations unchecked against claims; no record of searches or exclusions	Claim–source support verified, typed, and logged (Layer 3)
Deep-research agents	Breadth and report length	Coverage unmeasured; synthesis outruns evidence; hedges stripped	Register precedes prose; coverage scored against the charter (Layers 1–2)
Multi-agent pipelines	Throughput via role specialisation	Correlated agreement mistaken for verification; no independent challenge	Challenge is a separate adversarial function, measured against independent evidence (Layer 4)
Human-in-the-loop (nominal)	A reassurance label	Review unstructured, unrecorded, scaled to output volume — thins exactly when risk grows	Gates tiered by consequence, with named approvers and a review log (Layer 5)

The framework is deliberately *tool-agnostic*: it specifies functions, artefacts, and gates — not vendors, models, or architectures — because models change faster than governance should. Any agentic stack can be run inside it; and any stack that cannot produce the required artefacts (charter, source register, claim ledger, review log) is, by this paper's definition, not yet suitable for high-consequence research.

SECTION 5

5 The six-layer framework in depth

Six layers, four artefacts, one rule: nothing reaches a decision-maker that a named human cannot defend, claim by claim.

The framework is a pipeline with a spine. Work flows left to right — a question is framed, evidence is acquired and traced to claims, the emerging analysis is challenged, a human gates its release, and its life after release is monitored. The spine is human accountability: a person frames the question, a person can halt the pipeline at any layer, and a person signs the release. Each layer below is specified the same way — its purpose, how it works, the failure mode it exists to prevent, and the artefact it must produce. If a layer produces no artefact, it did not happen.

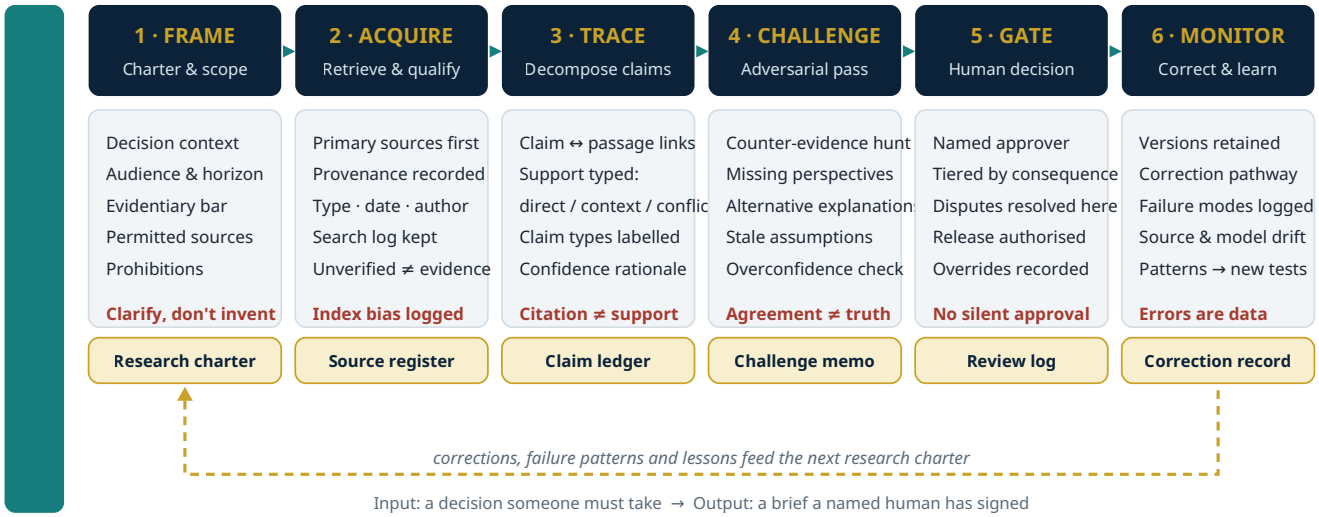


Figure 1 — The six-layer architecture. Work flows left to right; accountability runs vertically across every layer; monitoring feeds corrections back into future charters. Each layer's mandatory artefact is shown in gold. A layer that produces no artefact is treated as not having happened.

1 Question framing

THE CHARTER LAYER · HUMAN-LED

PURPOSE

To fix — before any model runs — what question is actually being asked, for whom, and what would count as a good answer. Most research failures are framing failures: the wrong question answered beautifully.

HOW IT WORKS

The research lead writes a short **research charter**: the decision the work informs; the intended audience; time horizon and geography; what counts as evidence; which source categories are required and which are permitted; which claims would be consequential if wrong; and explicit prohibitions (e.g., no claims about identifiable individuals without corroboration). The system must ask for clarification when the task is ambiguous rather than silently inventing scope, and the charter records the clarification exchange.

FAILURE MODE IT PREVENTS

Coverage illusion and scope drift — a comprehensive-looking answer to a question nobody asked, assembled from whatever sources were easiest to reach.

ARTEFACT

Research charter (versioned; approved at the first human gate).

2 Evidence acquisition

THE PROVENANCE LAYER

PURPOSE

To build the evidence base as a *register*, not a pile — every source qualified before it is allowed to support anything.

HOW IT WORKS

Retrieval prioritises primary and authoritative material. For each source the register records: type (official document, dataset, peer-reviewed work, journalism, commentary), author or institution, publication date, identifier or URL, retrieval date, and the role the source may play. Search terms and inclusion/exclusion decisions are logged so a reviewer can repeat or challenge the gathering process. Sources that cannot be accessed or verified are marked as such and may not count as confirmed evidence. Known retrieval biases — language, geography, index visibility — are logged as standing caveats.

FAILURE MODE IT PREVENTS

Fabrication by proxy (treating unverified material as established) and invisible coverage gaps.

ARTEFACT

Source register, including the search log and coverage caveats.

3 Claim-level traceability

THE AUDIT LAYER

PURPOSE

To make support a property of individual claims rather than a vibe of the whole document. This is the framework's centre of gravity.

HOW IT WORKS

Draft output is decomposed into **material claims** — assertions a reader might rely on. Each enters the **claim ledger** with: claim text; claim type (*fact, inference, forecast,* or *recommendation*); supporting source(s) with the exact passage or location; a support judgement — the source *directly supports* the claim, *provides context only*, or *conflicts* with it; a confidence rationale; and unresolved questions. Synthesis is written from the ledger, in a structure that keeps the five output categories visibly separate: established facts; contested or incomplete evidence; analytical interpretation; scenarios or forecasts (with assumptions and horizons stated); and recommendations (with the criteria used to compare options). A citation's presence is never treated as proof of support: the claim–source relationship is itself checked, by a component or reviewer other than the one that drafted the claim.

FAILURE MODE IT PREVENTS

False attribution and confidence laundering — the two failures readers are least able to detect unaided.

ARTEFACT

Claim ledger, from which the report's evidence map is generated.

4 Adversarial review

THE CHALLENGE LAYER

PURPOSE

To attack the draft before a decision-maker has to. A workflow without an institutionalised opponent will optimise for plausibility, because plausibility is what its components are rewarded for.

HOW IT WORKS

A separate pass — a differently-configured component, a second team, or a designated red-team reviewer — works from the claim ledger, not the prose. Its mandate: search for contradictory evidence and missing perspectives; construct the strongest alternative explanation for the same evidence; identify outdated assumptions and overconfident language; and test whether "consensus" claims reflect independent sources or one source repeated. Role separation in multi-agent designs can organise this work, but independence must be *designed and assessed*, not assumed from agent count: challenge quality is measured by the material issues it surfaces, and the challenge memo must be answered — accepted, rebutted with evidence, or carried into the report as stated disagreement — before the gate.

FAILURE MODE IT PREVENTS

Manufactured consensus and one-sided synthesis.

ARTEFACT

Challenge memo with dispositions for every material challenge.

5 Human decision gates

THE OWNERSHIP LAYER

PURPOSE

To ensure that consequential judgement is exercised by named people, at defined points, with the evidence in front of them — and that the record proves it.

HOW IT WORKS

Gates sit at charter approval, at claim review, and at release authorisation. Review intensity is tiered by consequence: low-risk background summaries may be sampled and source-checked; sensitive policy recommendations, or claims about identifiable people or organisations, require subject-matter expertise, may require legal review, or are excluded from automated processing altogether. Reviewers record what they checked, what they changed, what they overruled and why — **human-override outcomes are data**, feeding Layer 6. The system must never imply that an accountable human approved a conclusion when no such review occurred; an ungated output is labelled as such wherever it travels.

FAILURE MODE IT PREVENTS

Accountability evaporation and automation bias — the quiet slide from "AI-assisted" to "nobody checked".

ARTEFACT

Review log: gates, approvers, checks performed, overrides, release authorisation.

6 Monitoring & correction

THE LEARNING LAYER

PURPOSE

To make the workflow smarter after publication, not just before it — converting one-off quality control into an organisational learning loop.

HOW IT WORKS

Released outputs retain version information, source snapshots or identifiers where lawful, reviewer notes, and a standing correction pathway. When a material error is found — by the team, a reader, or events themselves — the failure mode is classified (against the atlas in Table 1), the output is corrected and re-versioned, sibling claims are checked for the same defect, and the lesson is converted into a test, a charter clause, or a retrieval rule. Monitoring also tracks drift: sources that change or disappear, shifts in model behaviour across versions, tool failures, and recurring error patterns, any of which can trigger re-review of affected outputs.

FAILURE MODE IT PREVENTS

Repeat errors and silent decay — the same mistake, made confidently, in every subsequent brief.

ARTEFACT

Correction record and failure-mode log, reviewed on a fixed cadence.

> **THE FOUR ARTEFACTS, TOGETHER**
>
> Charter, source register, claim ledger, review log. Together they let a third party reconstruct *why a brief says what it says* without interviewing its authors — which is the operational definition of auditable. They are also the pilot's raw material: Section 7's metrics are computed from these artefacts, not from impressions of the prose.

SECTION 6

6 A worked example: one question through all six layers

Abstract frameworks persuade nobody. Here is the pipeline doing a week's work on a single question. **ILLUSTRATIVE** *— the scenario, sources, and claims below are invented to show the mechanics; no such study has been run.*

The commission. A policy unit in a mid-sized emerging-market government is preparing advice on a five-year digital-infrastructure choice: invest in sovereign compute capacity, or subsidise access to foreign cloud AI services. A research team is asked for a brief in ten working days.

LAYER 1 — FRAME

The charter records: the decision (a cabinet-level investment direction, not procurement detail); the audience (senior officials, non-specialists); the horizon (2026–2031); consequential claims (anything about law, cost magnitudes, or national capability); required source categories (primary legal texts, official statistics, vendor material *flagged as interested*, independent technical analysis, at least two regional expert perspectives, relevant local-language coverage); and a prohibition — no cost figure may be stated without a dated, identified source.

LAYER 2 — ACQUIRE

Retrieval fills the register: the statute from the official gazette; two ministry procurement notices; national statistics on data-centre capacity; vendor white papers (role: interested party, context only for contested points); an independent analyst note; regional press in two languages. The search log records a gap the charter cares about — no public data on current public-sector cloud spend — logged as a declared evidence gap, not silently bridged.

LAYER 3 — TRACE

The draft is decomposed into 60+ claims; the ledger carries each one's support relationship. Three rows show the mechanics:

Table 4 — Worked example: three rows of a claim ledger. ***Illustrative*** *— invented for exposition.*

ID	Type	Claim	Support	Judgement	Confidence
C-014	Fact	The 2023 data-protection law requires public-sector health data to be stored in-country	Official gazette, statute §12 (primary)	DIRECT	High
C-021	Inference	Residency rules raise the effective cost of a foreign-cloud-only strategy for affected public workloads	C-014 + two ministry procurement notices	DIRECT (premises)	Medium — magnitude unquantified
C-027	Forecast	Regional accelerator (GPU) capacity remains supply-constrained through 2028	Vendor roadmap (interested) + one independent analyst note	CONTEXT	Low — presented as scenario, not finding

LAYER 4 — CHALLENGE

The challenge pass lands one material hit: C-021, as first drafted, generalised the residency rule to *all* public workloads; the statute covers health data and two other named categories. Disposition: the claim is narrowed ("for affected public workloads"), the inference's premises are re-checked, and the brief's executive summary is corrected to match. The challenger also flags that both cost sources trace to vendor ecosystems — recorded as a standing caveat in the brief.

LAYER 5 — GATE

Because the brief informs an investment direction, it sits in the highest review tier: the research lead reviews every high-consequence claim in the ledger; a legal reviewer confirms the statutory claims; the named approver signs the review log, including two overrides of machine-drafted language judged too definitive for the evidence. The overrides, with reasons, are logged — they become Layer 6 data.

LAYER 6 — MONITOR

The brief is versioned with its register and ledger. Three months later, a procurement notice contradicts one cost assumption: the correction pathway runs — affected claim identified from the ledger in minutes rather than re-researched from scratch, brief corrected to v1.1, failure mode classified (*stale source*), and a charter clause added for future work: cost claims require a source dated within twelve months.

WHAT THE EXAMPLE SHOWS

Nothing in this workflow required exotic technology. Its value is procedural: the error was caught by a *role* (the challenger), the correction was cheap because of an *artefact* (the ledger), and the lesson persisted because of a *loop* (monitoring into the next charter). Fluency was the input; auditability was the product.

SECTION 7

7 The evaluation protocol: metrics, baselines and blinded review

A pilot designed to be capable of embarrassing its designers. Thresholds are fixed before the run; failures are published with the scores.

A framework for auditable research must itself be auditable. This section specifies the evaluation the framework proposes to undergo — the same protocol outlined in summary in the original concept paper, here developed in full. It is a *test design*, not a report of results. Its logic: compare the framework-guided workflow against an honest baseline, on the same question, under the same conditions, scored by people who do not know which arm produced what, against thresholds fixed in advance.

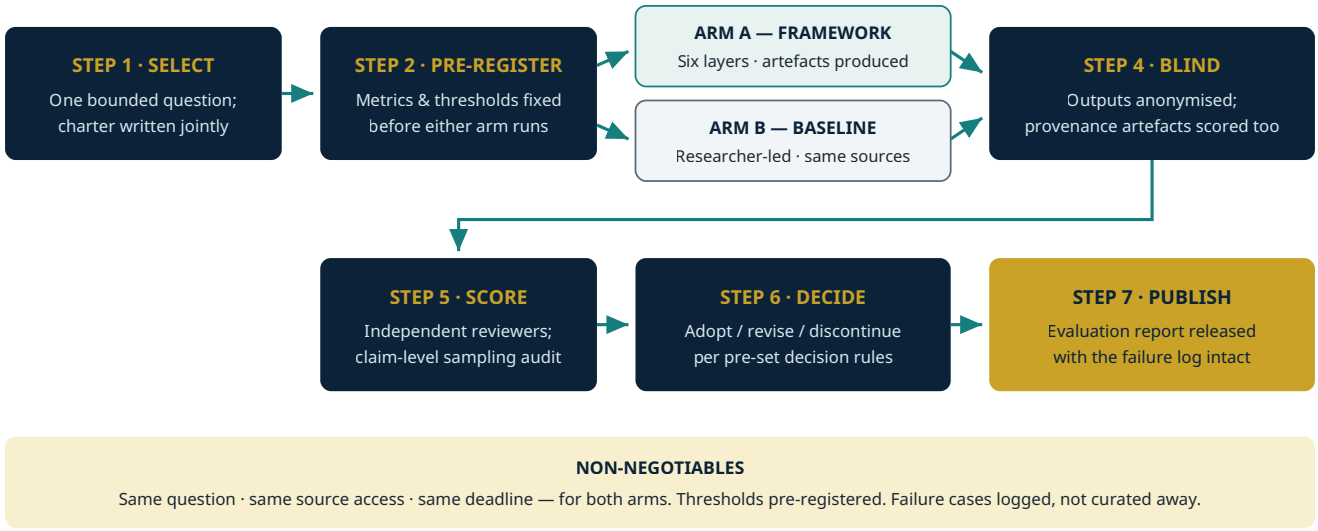


Figure 2 — The evaluation protocol. A two-arm, blinded comparison with pre-registered thresholds and a published failure log. The design's credibility rests on the non-negotiables: asymmetry between arms, or thresholds set after seeing results, would invalidate the exercise.

7.1 Metrics and proposed targets

Ten measures, computed from the framework's artefacts (register, ledger, review log) and from reviewer scoring. Targets are **TARGET** values proposed for a first pilot — design commitments to be confirmed with the pilot partner before the run, not findings and not universal constants.

Table 5 — Evaluation metrics. All thresholds are proposed targets for a first pilot, to be validated or revised — labelled TARGET throughout.

Metric	Definition	How computed	Proposed target
Sourced-claim rate	Share of material claims backed by at least one directly supporting, registered source	Audit of a random claim sample against the ledger	**TARGET** ≥ 95% overall; 100% for high-consequence claims
Citation precision	Share of citations whose cited passage actually supports the claim it is attached to	Independent checkers verify sampled claim–passage pairs	**TARGET** ≥ 90%
Citation recall	Share of material claims carrying the sources reviewers judge necessary to support them	Reviewer panel compares ledger against a gold source set for the question	**TARGET** ≥ 85%
Evidence coverage	Presence of the charter-mandated source categories and material counter-evidence	Charter checklist scored blind	

			TARGET 100% of mandated categories present, or the gap explicitly declared in the output
Fabrication rate	Share of sampled claims with no supporting source, or resting on an invented/misattributed reference	Ledger audit; every sampled reference resolved to a real source	**TARGET** ≤ 2% overall; zero invented references behind high-consequence claims
Calibration	Agreement between the confidence a claim states and its verified correctness	Claims grouped by stated confidence band; verified accuracy per band; Brier-style scoring	**TARGET** claims stated at ~90% confidence verify at ≥ 85%
Reproducibility	Share of the evidence trail an independent researcher can reconstruct from the artefacts alone	Reconstruction exercise by a researcher uninvolved in the run	**TARGET** ≥ 90% of material claims traceable end-to-end
Reviewer effort	Human minutes to verify, correct, and approve one brief to release standard	Time-logged gate reviews in both arms	**TARGET** ≤ baseline effort, at equal or better quality scores
Human-override outcomes	How often gate reviewers overrule machine-drafted content, in which direction, and whether events later vindicate the override	Review-log analysis, followed up at monitoring	**TARGET** Reported descriptively; a rising vindicated-override rate triggers a Layer 4 redesign review
Net time saved	Elapsed production time versus baseline, counted only when quality targets are also met	Pilot timing across both arms	**TARGET** Positive saving is a success *only if* the quality targets above are simultaneously met

7.2 Baselines: the comparison must be honest

The baseline is a competent conventional workflow — a researcher (or team) with the same question, source access, and deadline, using their usual tools, which may include ordinary LLM assistance without the framework's artefacts and gates. Stacking the comparison against a weak baseline would prove nothing; the question is whether the framework's overhead earns its keep against good existing practice. Both arms work from the jointly written charter, so the framework cannot win merely by having a better brief.

7.3 Blinded review: who scores, and how

Outputs from both arms are anonymised and format-normalised before scoring; reviewers do not know which arm produced which output, and arm assignment is held by a coordinator who does no scoring. Reviewers score the prose *and* the audit trail: the framework arm submits its artefacts; the baseline arm submits its notes and sources in whatever form it naturally produced them, and the reproducibility metric measures exactly that difference. Claim scoring uses stratified random sampling — high-consequence claims always at 100% — with two independent checkers per claim and adjudication of disagreements. Reviewer agreement is reported; a metric whose checkers cannot agree is a finding about the metric, published as such.

7.4 Decision rules: what the scores would mean

- **Adopt and extend** — quality targets met, reviewer effort at or below baseline: the framework earns a second, larger pilot on a harder question.
- **Revise** — quality targets met but effort materially above baseline, or specific layers show weak marginal value: simplify or re-tier those layers and re-test. A framework whose safety costs more than the risk it removes will be abandoned by its users, and should be.
- **Discontinue** — fabrication or coverage targets missed, or calibration worse than baseline: the workflow, as designed, does not do the thing it exists to do. The evaluation report says so, in those words.

Failure cases are logged throughout in the atlas categories of Table 1 — false citations, omitted evidence, duplicated sources presented as independent, prompt-injection attempts in retrieved material, privacy near-misses, and observed automation bias at the gates — and the log is published with the evaluation, not summarised away.

SECTION 8

8 Pilot-study design for a research partner

One bounded topic, one honest baseline, one published evaluation. A pilot is a question, not a demonstration.

The framework's next step is a bounded pilot with a research partner — an institution with real strategic questions and the standing to judge whether the answers improved. The design below develops the pilot proposed in the concept paper, structured so that a negative result is as publishable, and as useful, as a positive one.

8.1 Shape of the pilot

- **Topic.** One bounded question in AI governance, emerging technology, or public policy — consequential enough that quality matters, bounded enough that a gold source set and claim audit are feasible.
- **Stage 1 — Agree.** Research question, audience, source boundaries, review tiers, thresholds, and the baseline design are fixed jointly and recorded in the charter. Data rules: public or otherwise authorised data only; no sensitive personal information; human approval gate before any external publication.
- **Stage 2 — Produce.** Both arms run (Section 7). The framework arm produces the brief plus its four artefacts; the baseline arm produces its brief plus its natural working record.
- **Stage 3 — Verify.** Claim-level verification and adversarial review inside the framework arm; blinded scoring across both arms by reviewers independent of production.
- **Stage 4 — Evaluate.** Scores against pre-registered thresholds; failure log compiled; adopt / revise / discontinue recommendation written against the decision rules of §7.4.

8.2 Deliverables

A concise strategic brief on the pilot question; the source register; the claim ledger; the review and correction log; a short evaluation report including the failure log; and recommendations on whether the workflow should be revised, repeated, or discontinued. Deliverables belong to the evaluation as much as to the topic: the partner receives not just an answer, but the means to judge the answer.

8.3 Terms that protect both sides

A pilot does not require the partner to endorse Oryx, the framework, or any unvalidated claim. Any public case study, logo use, or statement about collaboration requires prior written consent from all relevant parties. Scope, ownership, confidentiality, security, and any commercial terms are agreed in writing before work begins. If the evaluation is negative, that fact — not a softened version of it — is what gets reported. These terms are not boilerplate; for a governance framework, they are the product being demonstrated.

8.4 Implementation note (October 2026): what has been built since v2.0

Since v2.0, the framework has been implemented as an internal minimum viable product: a nine-stage pipeline (framing, evidence acquisition, screening, synthesis, structured evidence-table construction, mechanical verification, adversarial review, human gate, PDF export with a page-1 Audit Certificate) that maps onto the six layers, operated through a local web interface. In October 2026 the implementation completed a first end-to-end verification run on a single test question — recent evidence on AI adoption in Indian healthcare, 2024–2026 — in 40.9 seconds: MEASURED 10 candidate sources screened, 3 included, 13 verification markers checked, none invalid, verification PASS, and a sourced-claim rate of 36% (mechanical sentence rule: a sentence counts as cited if it contains at least one valid source marker). The run used a single model (gemini-3.1-flash-lite), ran on localhost only with no public deployment, and the human gate was exercised in test form only — not by an independent reviewer. This run verifies that the pipeline executes and produces its artefacts; it does not validate the framework's performance, which only the blinded partner pilot of §7–§8 can do. The protocol, targets, and pilot design above are unchanged.

SECTION 9

9 The Global South dimension

Where evidence is hardest to gather is exactly where unverifiable synthesis does the most damage — and where auditable method has the most to offer.

Strategic research *about* the Global South faces a representational challenge that the auditability gap sharpens. Relevant evidence is distributed across languages, institutions, local media, policy documents, and data systems with unequal digital visibility. Retrieval weighted toward highly indexed English-language sources can reproduce an incomplete picture while appearing comprehensive — and an agentic system will then synthesise it fluently, at scale, with citations. This is a design risk to engineer against, not a claim that all current systems behave identically.

The framework responds at three points. **At the charter (Layer 1)**, language and geographic coverage are explicit design choices: which languages will be searched, which local source categories are mandatory, and which gaps are already known to exist. **At acquisition (Layer 2)**, the register logs coverage as it actually happened — sources sought versus sources obtained, by language and origin — so that a brief's evidence base can be described honestly: "strong on official English-language material; thin on provincial implementation reporting" is a finding, and belongs in the brief. **At the gate (Layer 5)**, where interpretations depend on institutional or cultural context, local subject-matter review is a named requirement, not a courtesy. The objective is not to treat any region as homogeneous, but to make visible whose evidence enters a strategic analysis and whose perspectives may be missing.

There is also a capacity argument, and it cuts both ways. For institutions with limited research staff, agentic tools may organise large source collections and remove repetitive work that currently crowds out analysis. But deployment must be proportionate to governance capacity: a team needs data-handling rules, a way to verify outputs, staff training, and an escalation path for high-risk findings. A low-cost tool that cannot be audited creates hidden costs — errors, reputational harm, misplaced confidence — that fall hardest on institutions least able to absorb them. The framework's artefacts are deliberately low-technology: a charter, a register, a ledger, and a log can be kept in ordinary documents by a small team. Auditability should be a property of method, available to a five-person institute in Delhi or Nairobi on the same terms as to a large Northern agency — not a premium feature of expensive platforms.

SECTION 10

10 Risks and safeguards

Six standing risks, each paired with the safeguard this framework treats as mandatory — adapted to task and jurisdiction, never applied as a thoughtless checklist.

Table 6 — Risk register. Consistent with the direction of NIST risk management, OECD principles, and UNESCO's human-centred guidance.

Risk	How it arises	Mandatory safeguard
Hallucination & false attribution	Generated claims unsupported; citations incorrect or irrelevant	Claim-level source checks; generated references are unverified until inspected (Layer 3)
Source bias & coverage gaps	Retrieval over-represents well-indexed institutions and familiar viewpoints	Predefined source categories; search coverage logged; counter-evidence actively sought; gaps declared (Layers 1–2)
Correlated multi-agent error	Agents share model, data, and assumptions; consensus is manufactured	Diversify sources and review methods; measure agreement against independent evidence, never by counting agent consensus (Layer 4)
Prompt injection & tool misuse	Retrieved content contains instructions designed to redirect the system	Retrieved content treated as untrusted data; instructions isolated from source text; tool permissions constrained; consequential actions require human approval (Layers 2, 5)
Privacy, confidentiality & IP	Research material includes restricted data or copyrighted content	Authorised inputs only; sensitive data minimised; licence terms respected; legal or organisational review where required (Layers 1, 5)

Automation bias & accountability gaps	Reviewers over-trust fluent output or assume another reviewer checked it	Named accountable reviewer; checks recorded; uncertainty and dissent shown, not smoothed; gates explicit (Layer 5)

SECTION 11

11 An 18-month roadmap

From this concept paper to a first independently reviewed pilot — in four phases, each with an exit test. These are plans and intentions, stated as such; they describe no existing commitments, partnerships, or funding.

Status note (October 2026). Phase 1's framework publication is done (v2.0, Zenodo); Phase 2 implementation has begun with the internal MVP verification run described in §8.4. Phases 3 and 4 — the partner pilot and its published evaluation — remain ahead, and no partner, funding, or commitment is claimed for them.

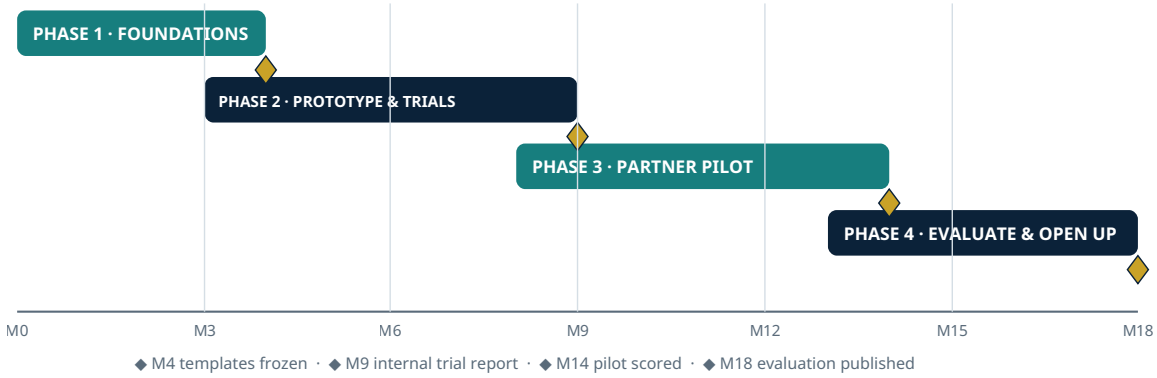


Figure 3 — The 18-month roadmap. Overlapping phases with milestone gates (gold diamonds). Each phase begins only when the previous phase's exit test (Table 7) is met; the roadmap is a plan, and slippage will be reported as slippage.

Table 7 — Roadmap phases: objectives and exit tests. All entries are intentions, not commitments.

Phase	Objectives	Exit test
1 · Foundations Months 0–4	Publish this framework for open comment; build the four artefact templates (charter, register, ledger, review log); write data-handling, privacy, and consent rules; recruit an informal external reading circle for critique	Templates frozen at v1.0; governance rules documented; substantive external critique received and answered
2 · Prototype & internal trials Months 3–9	Implement the workflow on available tooling; run internal dry-run studies on historical questions with known answers; build the failure-mode atlas from observed errors; calibrate review tiers	Internal trial report completed, including at least one documented failure the framework caught — and an honest account of any it missed
3 · Partner pilot Months 8–14	Agree and run the Section 8 pilot with a research partner: two arms, blinded scoring, pre-registered thresholds	Pilot scored against every metric in Table 5; failure log complete
4 · Evaluate & open up Months 13–18	Publish the evaluation — scores, failure log, and the adopt / revise / discontinue recommendation; release templates and protocol openly for reuse and challenge; design a second-round study informed by the results	Evaluation publicly released; framework v2.0 issued, revised in light of evidence — or discontinued with reasons stated

SECTION 12

12 Limitations, stated plainly

A governance paper that hides its own weaknesses would refute itself. These are ours.

- **The framework's performance has not been tested.** The framework is a design hypothesis synthesised from literature and guidance. No element of it has been validated by a blinded pilot, experiment, or peer review, and the proposed targets in Table 5 are judgement calls awaiting calibration — several may prove unachievable, or trivially achievable, in practice. What has been measured since v2.0 is the problem, not the solution: the companion citation-audit study (§2) and a single internal implementation verification run (§8.4). Neither substitutes for the §7 pilot.
- **The verification run is narrow.** One test question, one model, one internal run, localhost only, with the human gate exercised in test form. It demonstrates that the pipeline executes and produces its artefacts; it says nothing yet about performance across questions, models, reviewers, or real partner conditions.
- **The synthesis is the authors'.** A narrative review reflects selection judgement. Other readers of the same literature would weight the failure modes and the safeguards differently; the framework should be read as an argument, not a consensus.
- **Overhead is real, and may not pay.** Charters, registers, ledgers, and gates cost skilled human time. For many research tasks — genuinely low-stakes ones — that cost will exceed the risk retired. The framework is tiered for this reason, but the tiers themselves are untested judgement.
- **Traceability is not truth.** A claim can be perfectly traced to a source that is itself wrong, captured, or misread at the passage level. The framework audits the *relationship* between claims and evidence; it cannot certify the evidence. Source criticism remains a human scholarly duty the workflow supports but does not discharge.
- **Metrics can be gamed.** Once targets exist, workflows will optimise toward them — fuller ledgers, safer claims, narrower questions. The evaluation design mitigates this (blinding, gold sets, failure logs) but does not eliminate it; Goodhart's law applies to governance frameworks too.
- **Small-team authorship.** Oryx is an early-stage, two-founder initiative, not yet formally registered, writing from Delhi. The paper's Global South discussion is a design perspective offered for testing and critique — not a claim to represent the research institutions of the South, whose practices and constraints vary enormously.

SECTION 13

13 Conclusion

Credibility in the age of fluent machines belongs to whoever can show their working.

Agentic AI will make research workflows more scalable; that much is already visible. The open question is whether it will make them more *trustworthy* — and that depends on design choices being made now, mostly by default. The default design optimises for answers: faster, longer, smoother. This paper has argued for a different optimisation target. Strategic research needs the relationship between evidence, claims, uncertainty, and accountable judgement to survive automation — and survival, here, is an engineering and governance property, not a hope.

The six-layer framework — frame, acquire, trace, challenge, gate, monitor — is our attempt to specify that property operationally, and the evaluation protocol is our attempt to keep ourselves honest about whether it works. Both are offered in the spirit the framework demands: as claims with their reasoning visible, open to challenge, and subordinate to evidence we have not yet gathered. The next step is a bounded pilot with transparent criteria and a documented baseline. If the workflow improves the auditability and usefulness of research without shifting hidden verification costs onto reviewers, it may give research institutions — including small ones, including those far from the centres of AI development — a practical foundation for using agentic AI responsibly. If it does not, the evaluation should make that visible, and this paper will have served its purpose anyway: in this domain, credibility depends not on claiming that AI can replace judgement, but on designing systems that help people exercise judgement better — and on being willing to publish what the test says.

“Do not ask whether the machine's answer sounds right. Ask whether you can show, claim by claim, why it should be believed — and who has agreed to stand behind it.”

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Author note and status

Amartya Kumar is the Founder of the Oryx Centre for Strategic Futures; **Riya Nalwa** is its Co-Founder. Oryx is an early-stage, non-profit research and policy initiative based in New Delhi, India, focused on artificial intelligence, geopolitics, emerging technologies, strategic foresight, and the future of the Global South. The Centre is not yet formally registered in any jurisdiction. This working paper was prepared independently for discussion and feedback. It is not a peer-reviewed publication; it reports no blinded-pilot results, no publicly deployed product, no funding, and no institutional endorsement. Empirical results cited from the companion study (§2) and the internal verification run (§8.4) are labelled as such where they appear. Proposals in it — including the pilot of Section 8 and the roadmap of Section 11 — are proposals: they describe work the authors intend to seek the partners, scrutiny, and support to carry out.

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Suggested citation: Kumar, A., & Nalwa, R. (2026). *From Answers to Auditable Intelligence: A Governance-First Framework for Agentic AI in Strategic Research and Public Policy* (Working Paper v3.0). Oryx Centre for Strategic Futures, New Delhi.



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The Oryx Centre for Strategic Futures studies how artificial intelligence is reshaping research, policy, and statecraft — and builds methods for using it accountably. Our work is methodological and empirical in intent: frameworks, evaluations, and open publications that other institutions can test, reuse, and challenge.

We welcome conversation with research institutions, funders, technology-assessment bodies, and public agencies working on the governance of advanced AI — particularly those interested in subjecting this framework to the test it asks for.

Frame · Acquire · Trace · Challenge · Gate · Monitor

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